

REACT NATIVE NAVIGATION

BUILDING
A NAVIGATION-BASED APP

DEMO





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REACT NATIVE NAVIGATION

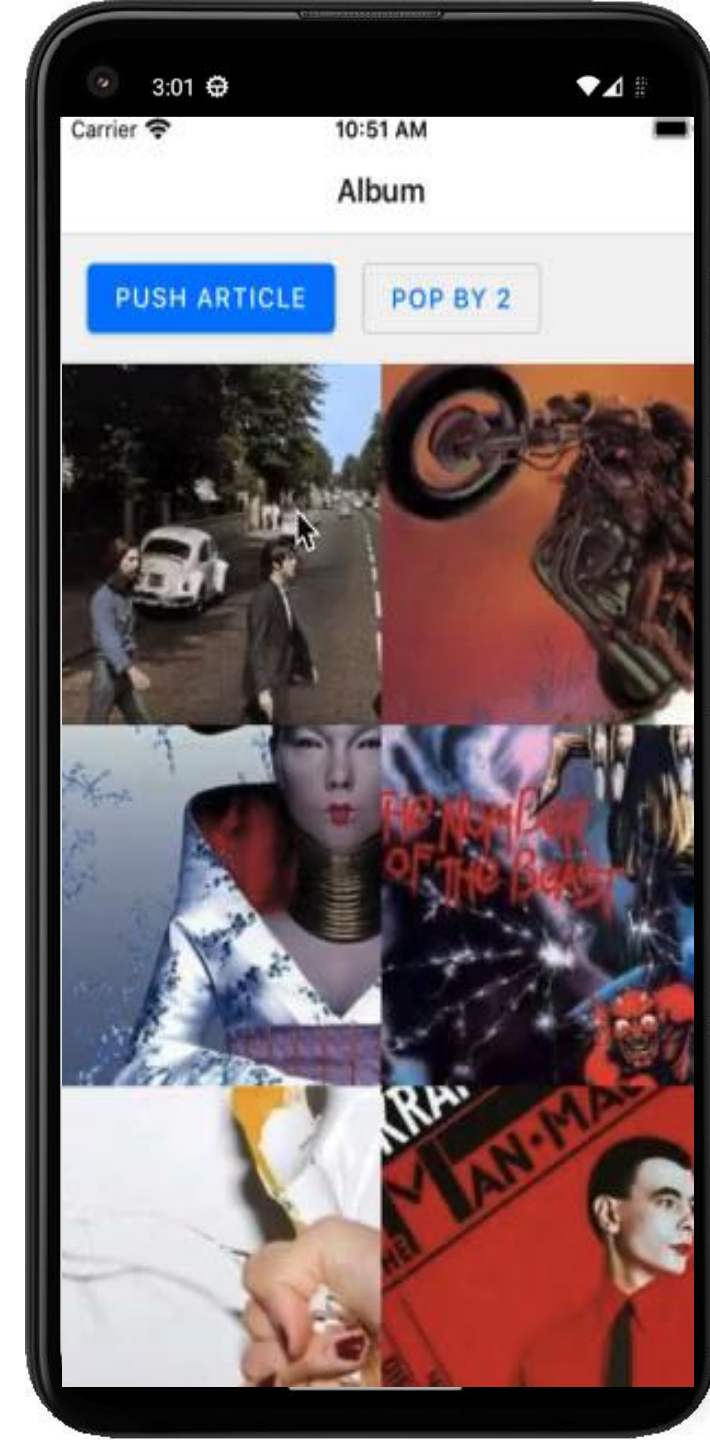
React Native Navigation vs. Web Navigation

- In web application
 - Working with URLs
- In mobile application
 - Stack-based
 - Tab-based
 - Drawer-based

Stack Based Navigation

- Transition between screens where each new screen is placed on top of a stack
- In Stack based navigation
 - Moving forward
 - Go backward

View Demo



Tab Based Navigation

- Pressing a tab takes to the screen that correlates with the tab
- Tab navigation
 - Top
 - Bottom

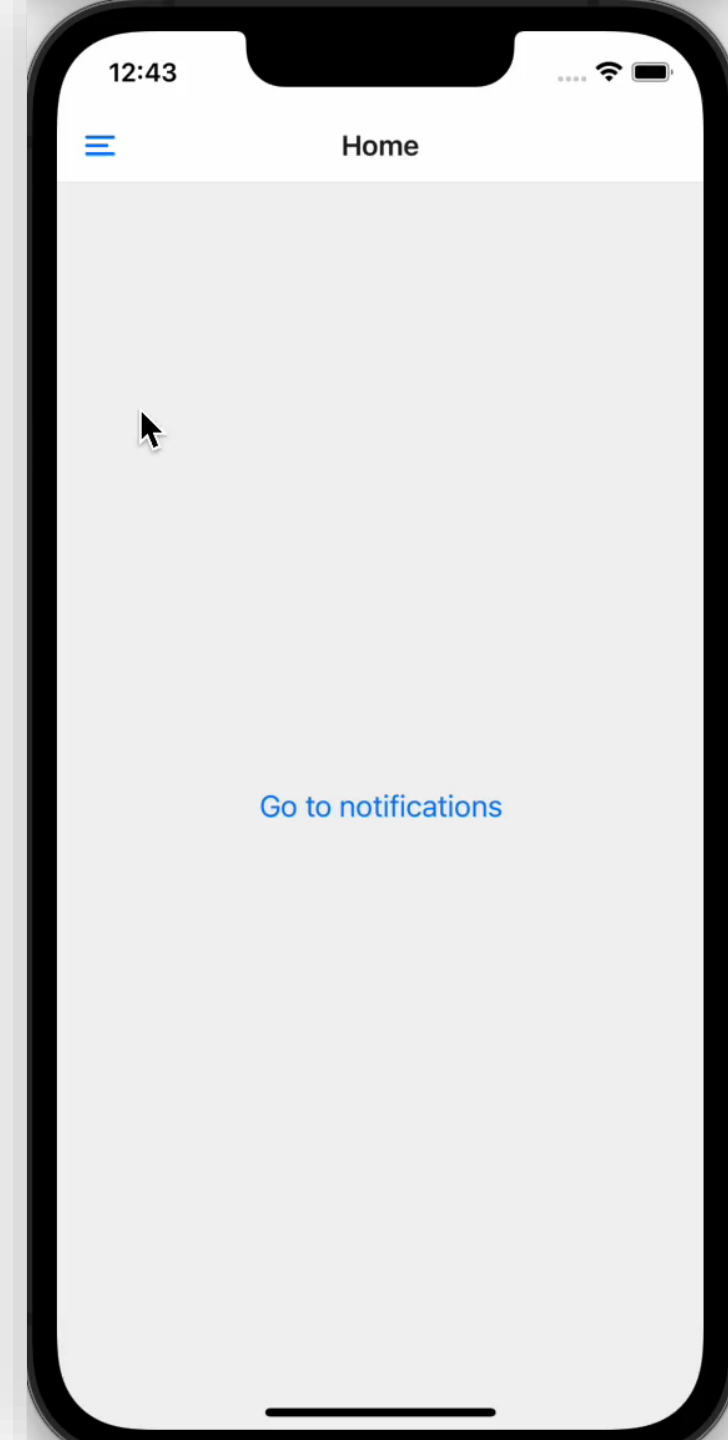
View Demo



Drawer Based Navigation

- Typically a side menu that pops out from either the left or right side of the screen and shows a list of options
- When you press an option, the drawer closes, and you're taken to the new screen

View Demo





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BUILDING A NAVIGATION-BASED APP

Setup The Navigation

- **NavigationContainer** - responsible for managing your app state and linking your top-level navigator to the app environment
- **Navigator** - managing the presentation of, and transition between, multiple screens
- **Screen** components - are used to configure various aspects of screens inside a navigator

Setup Navigation

- Install navigation library
 - `npm install @react-navigation/native`
 - `npx expo install react-native-screens react-native-safe-area-context`
- Install navigator
 - `npm install @react-navigation/native-stack`
 - `npm install @react-navigation/bottom-tabs`
 - `npm install @react-navigation/drawer`
 - `npm install react-native-gesture-handler react-native-reanimated`

Stack Navigation Sample

- Library

`npm install @react-navigation/native-stack`

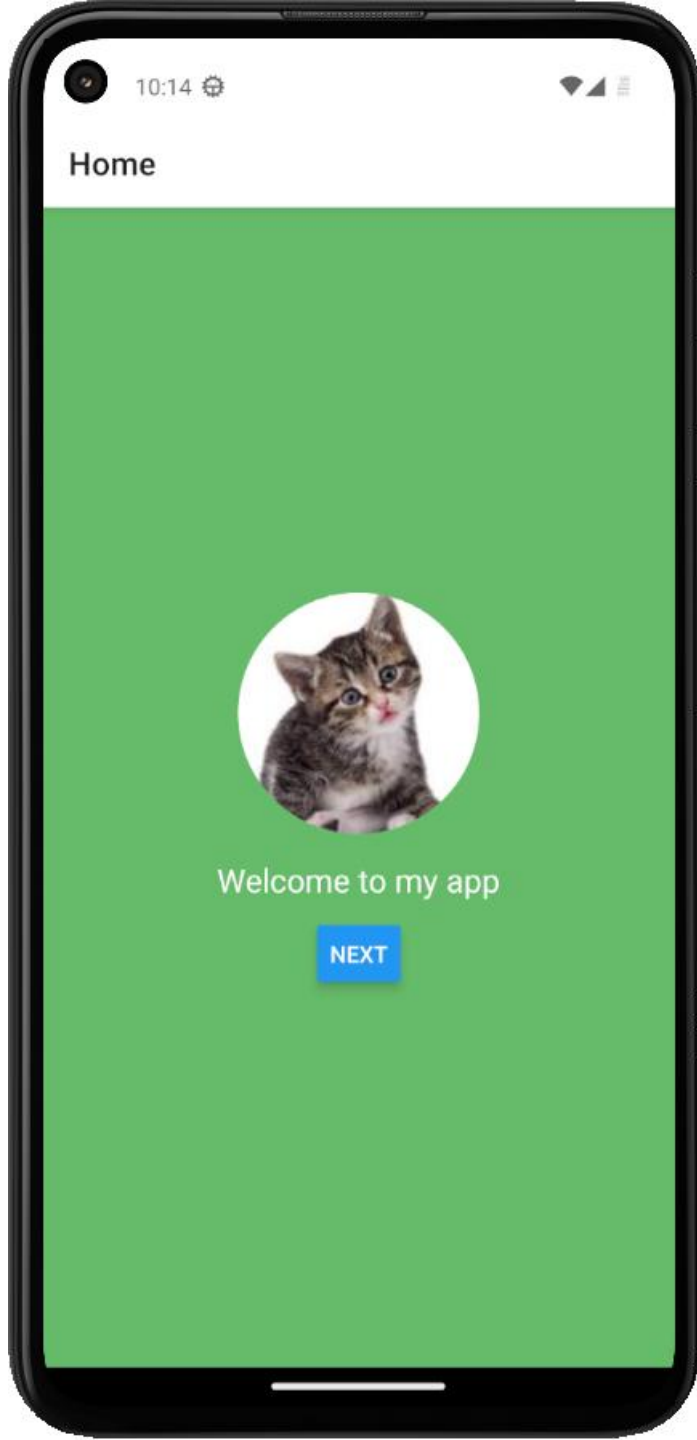
- Create Stack navigator

```
const Stack = createNativeStackNavigator();
```

Moving Between Screens

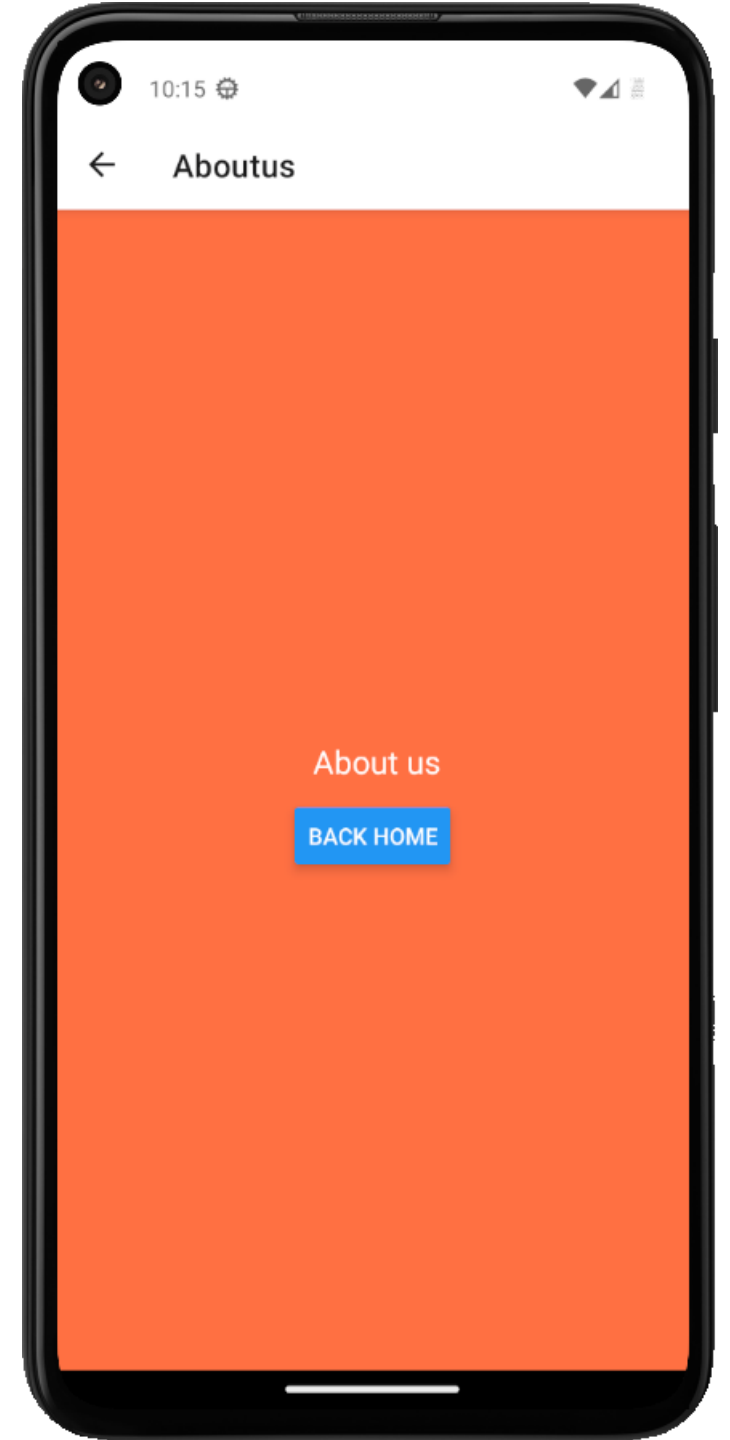
navigation

- `navigate` - go to another screen, figures out the action it needs to take to do it
- `reset` - wipe the navigator state and replace it with a new route
- `goBack` - close active screen and move back in the stack
- `setParams` - make changes to route's params
- `dispatch` - send an action object to update the [navigation state](#)
- `setOptions` - update the screen's options
- `isFocused` - check whether the screen is focused
- `addListener` - subscribe to updates to events from the navigators

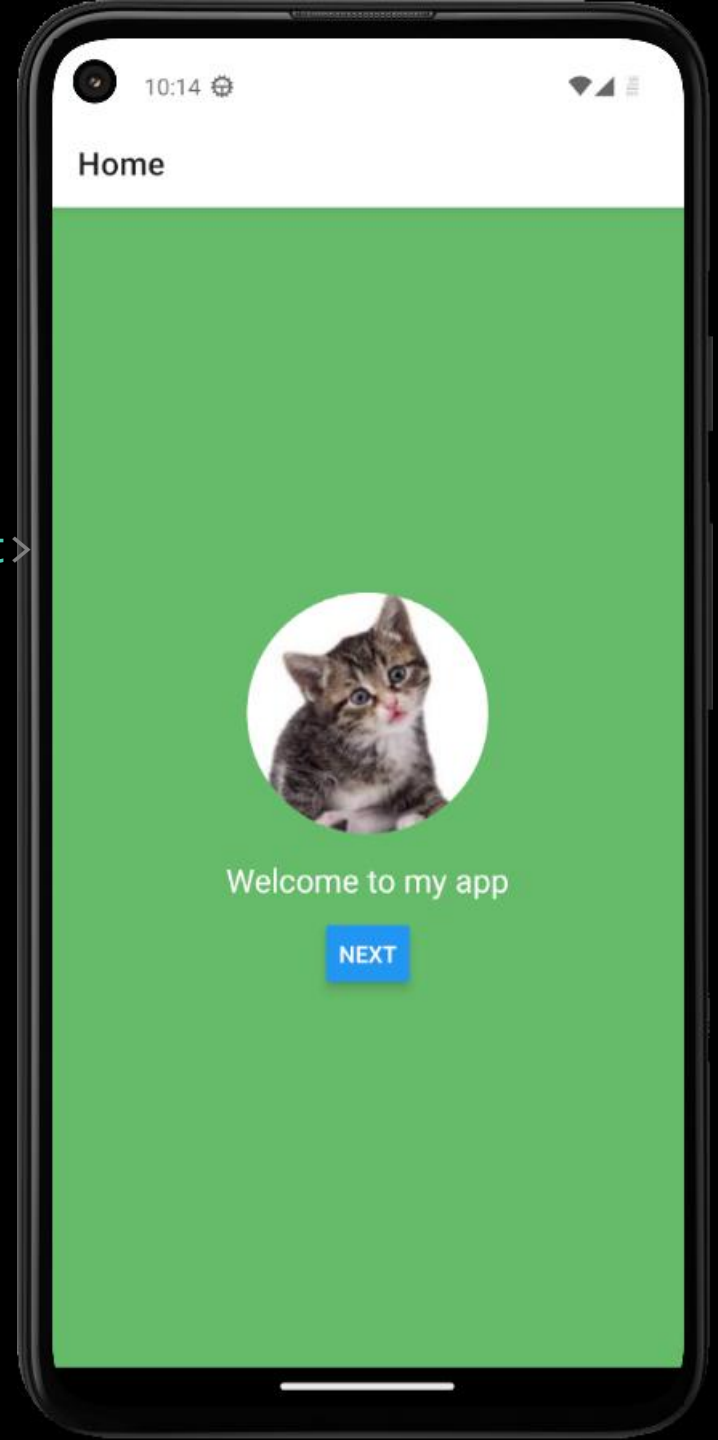


```
onPress={()=>  
  navigation.navigate("Home")}
```

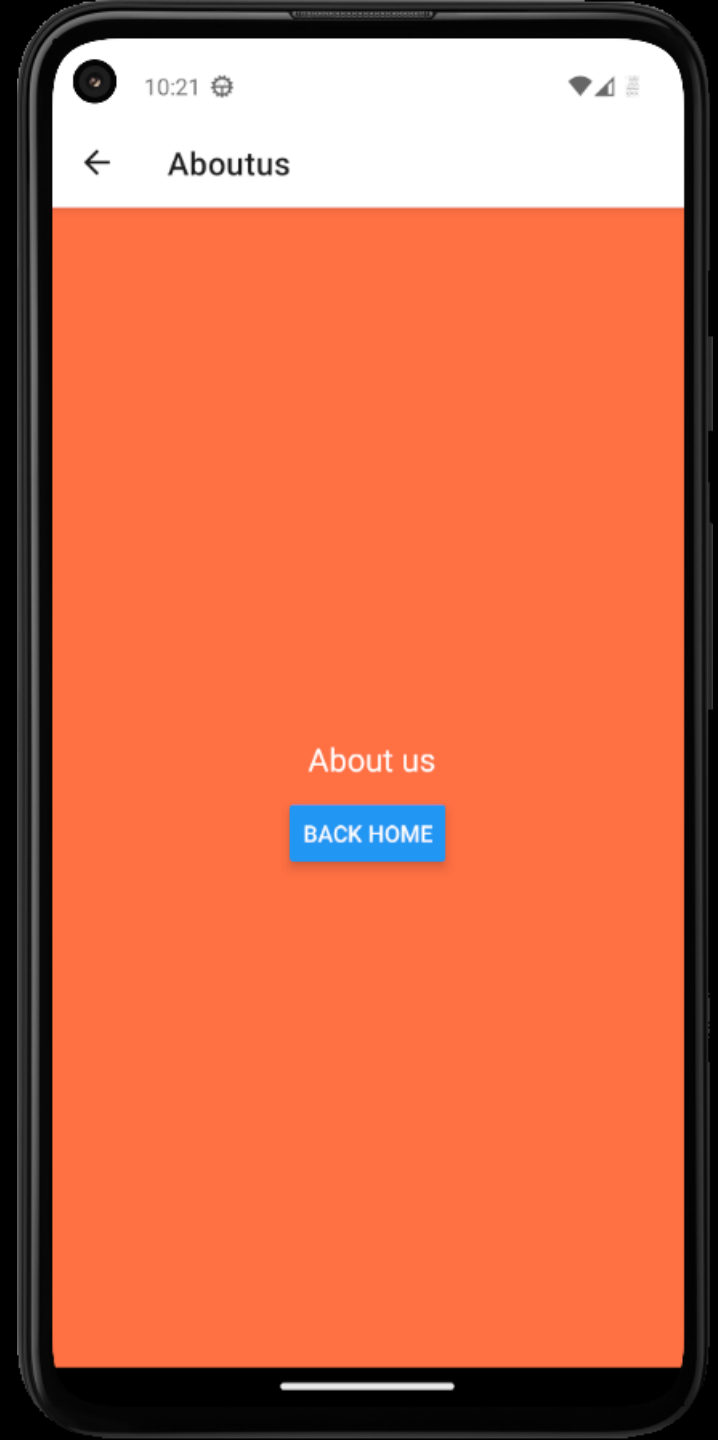
```
onPress={ ()=>  
  navigation.navigate("Aboutus") }
```



```
const Screen1 = ({navigation}) => {  
  return (  
    <View style={styles.screen1}>  
      <Image source={require("../assets/img1.jpg")}  
        style={styles.img}/>  
      <Text style={styles.title}> Welcome to my app </Text>  
      <Button  
        title="Next"  
        onPress={() => navigation.navigate("Aboutus")} />  
    </View>  
  );  
}
```



```
const Screen2 = ({navigation}) =>{  
  return (  
    <View style={styles.screen2}>  
      <Text style={styles.title}> About us</Text>  
      <Button  
        title="Back home"  
        onPress={() => navigation.navigate("Home")} />  
    </View>  
  );  
}
```



```
import { NavigationContainer } from '@react-navigation/native';
import { createNativeStackNavigator } from '@react-navigation/native-stack';

const Stack = createNativeStackNavigator();
const StackNav = () =>
{
  return (
    <NavigationContainer>
      <Stack.Navigator>
        <Stack.Screen
          name="Home"
          component={Screen1}
          options={{ title: 'Home' }}
        />
        <Stack.Screen
          name="Aboutus"
          component={Screen2}
          options={{ title: 'Aboutus' }} />
      </Stack.Navigator>
    </NavigationContainer>
  )
}
```

```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: '#000',
    alignItems: 'center',
    justifyContent: 'center',
    height: '100%',
    width: '100%'
  },
  img: {
    width: 150,
    height: 150,
    borderRadius: 150/2
  },
  title : {
    fontSize: 20,
    fontWeight: 'normal',
    color: '#fff',
    paddingTop: 15,
    paddingBottom: 15
  },
},
```

```
screen1: {
  flex: 1,

  backgroundColor: '#66bb6a',
  alignItems: 'center',
  justifyContent: 'center',
  height: '100%',
  width: '100%'
},
screen2: {
  flex: 1, backgroundColor: '#ff7043',
  alignItems: 'center',
  justifyContent: 'center',
  height: '100%',
  width: '100%'
},
screen3: {
  flex: 1,
  backgroundColor: '#f06292',
  alignItems: 'center',
  justifyContent: 'center',
  height: '100%',
  width: '100%'
}
});
```

Passing Data To Another Screen

route

- **key** - Unique key of the screen. Created automatically or added while navigating to this screen.
- **name** - Name of the screen. Defined in navigator component hierarchy.
- **path** - An optional string containing the path that opened the screen, exists when the screen was opened via a deep link.
- **params** - An optional object containing params which is defined while navigating e.g.

Home Screen

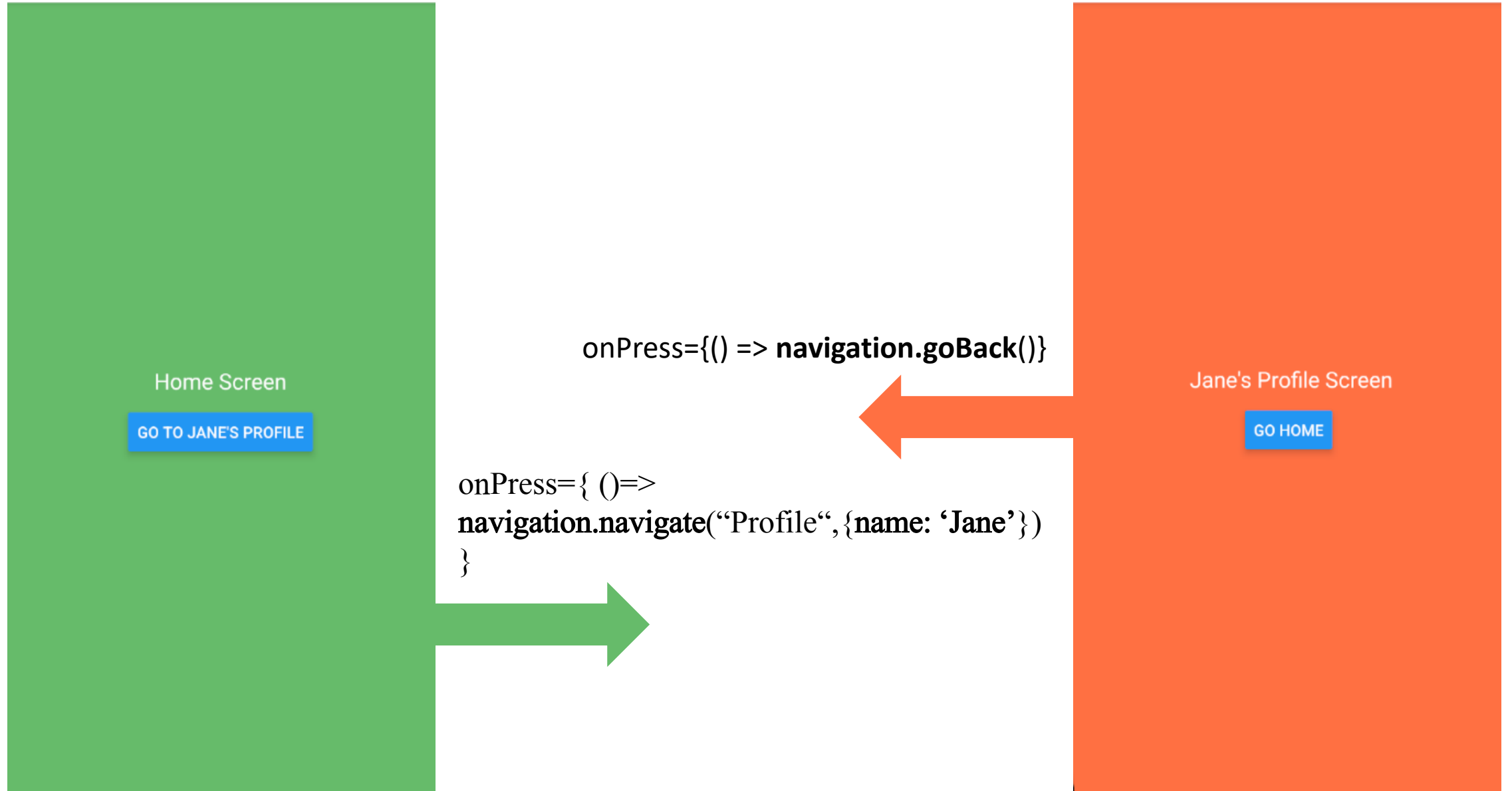
GO TO JANE'S PROFILE

`onPress={() => navigation.goBack()}`

`onPress={ ()=>
navigation.navigate("Profile",{name: 'Jane'})
}`

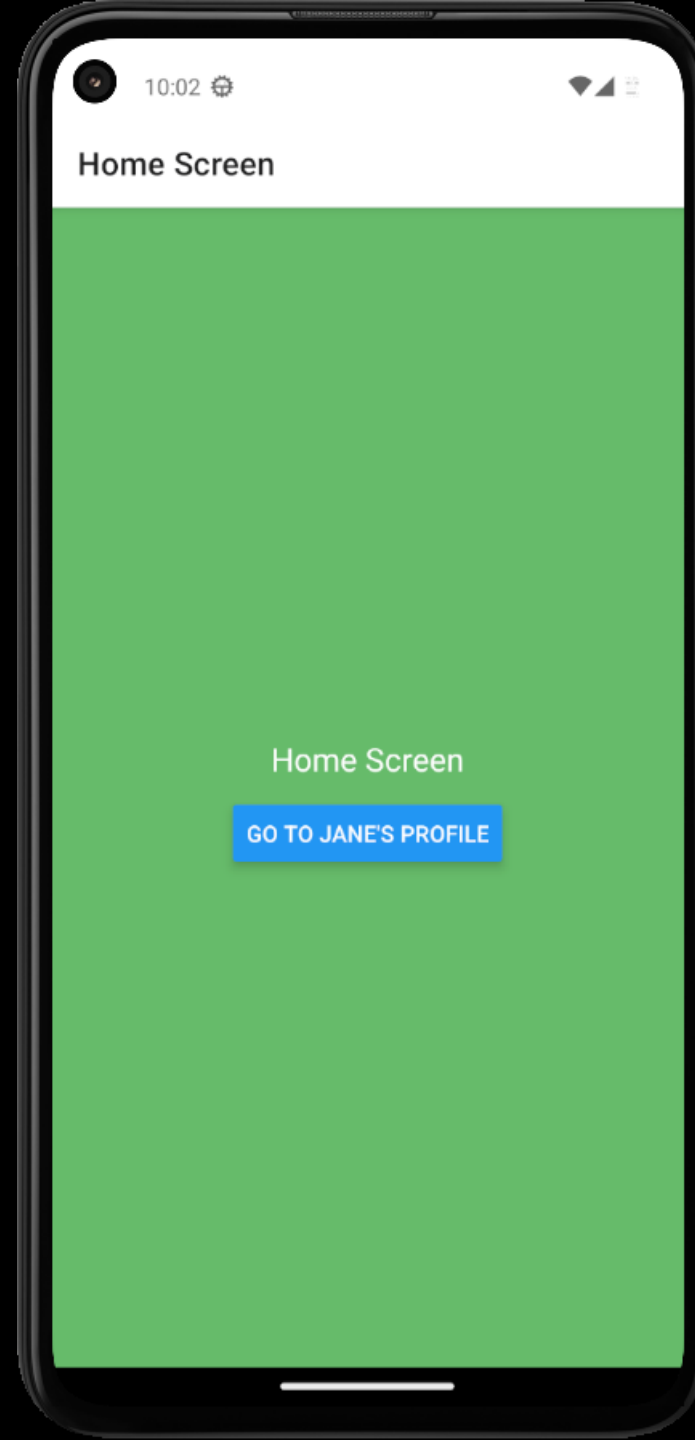
Jane's Profile Screen

GO HOME

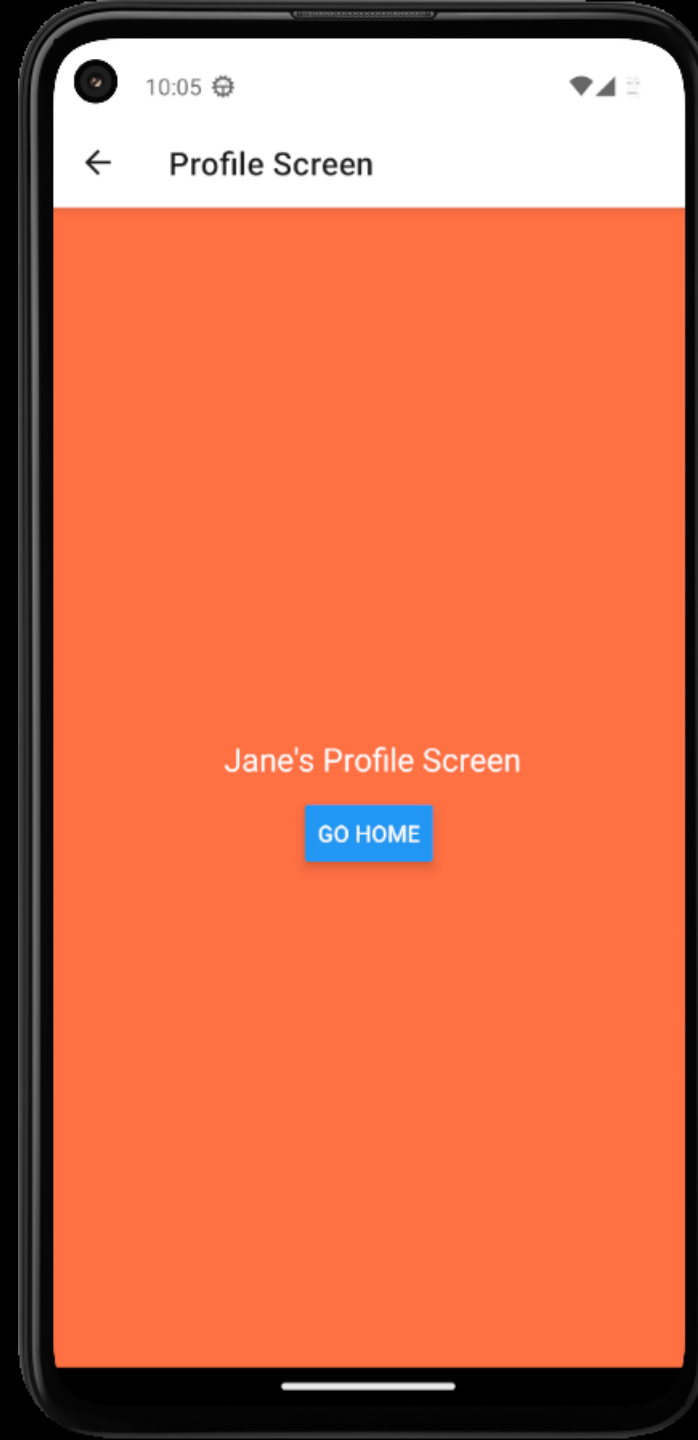


```
const Stack = createNativeStackNavigator();
const MyStack = () =>
{
  return (
    <NavigationContainer>
      <Stack.Navigator>
        <Stack.Screen
          name="Home"
          component={HomeScreen}
          options={{ title: 'Home Screen' }}
        />
        <Stack.Screen
          name="Profile"
          component={ProfileScreen}
          options={{ title: 'Profile Screen' }} />
      </Stack.Navigator>
    </NavigationContainer>
  )
}
```

```
const HomeScreen = ({navigation}) => {  
  return (  
    <View style={styles.screen1}>  
      <Text style={styles.title}> Home Screen </Text>  
      <Button  
        title="Go to Jane's profile"  
        onPress={() =>  
          navigation.navigate("Profile",{name:'Jane'})} />  
    </View>  
  );  
}
```



```
const ProfileScreen = ({navigation, route}) =>{  
  return (  
    <View style={styles.screen2}>  
      <Text style={styles.title}>  
        {route.params.name}'s Profile Screen  
      </Text>  
  
      <Button  
        title="GO HOME"  
        onPress={() => navigation.goBack()} />  
    </View>  
  );  
}
```



Tab Navigation Sample

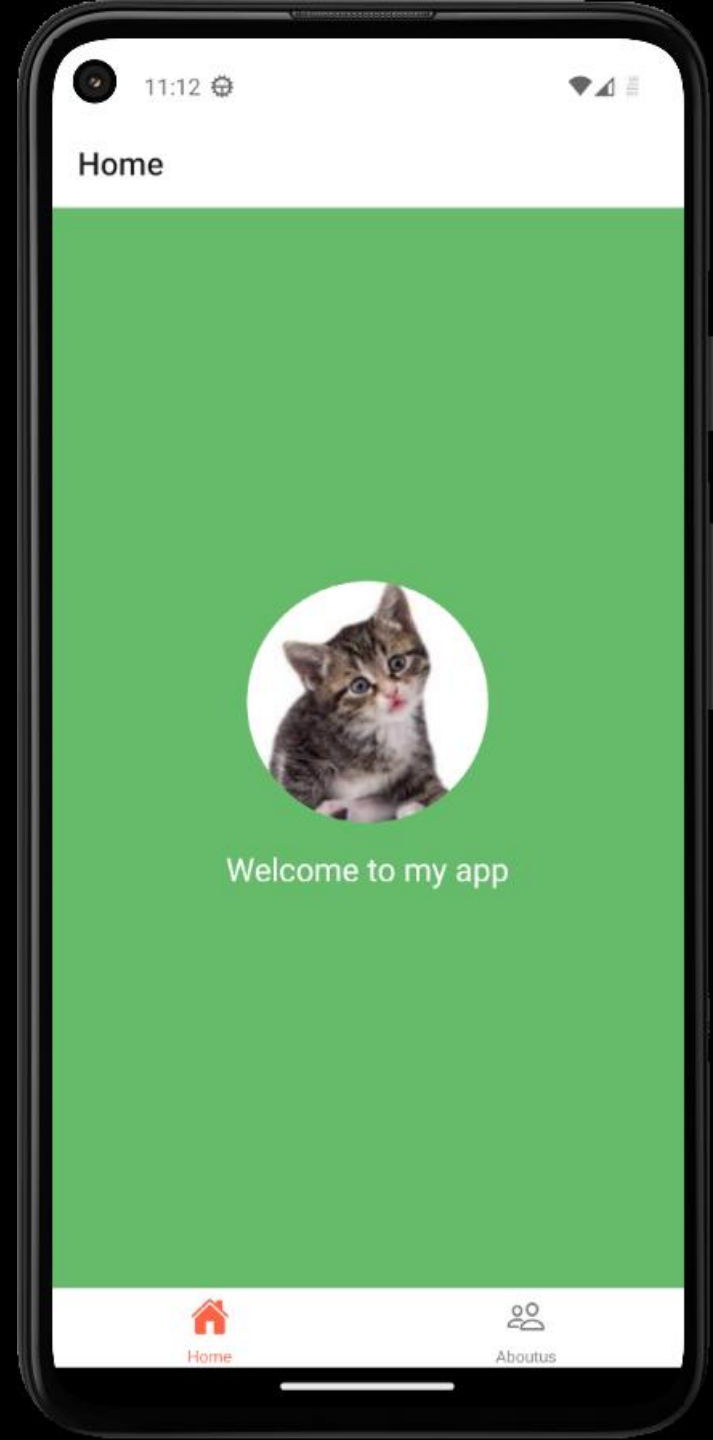
- Library

```
npm install @react-navigation/bottom-tabs
```

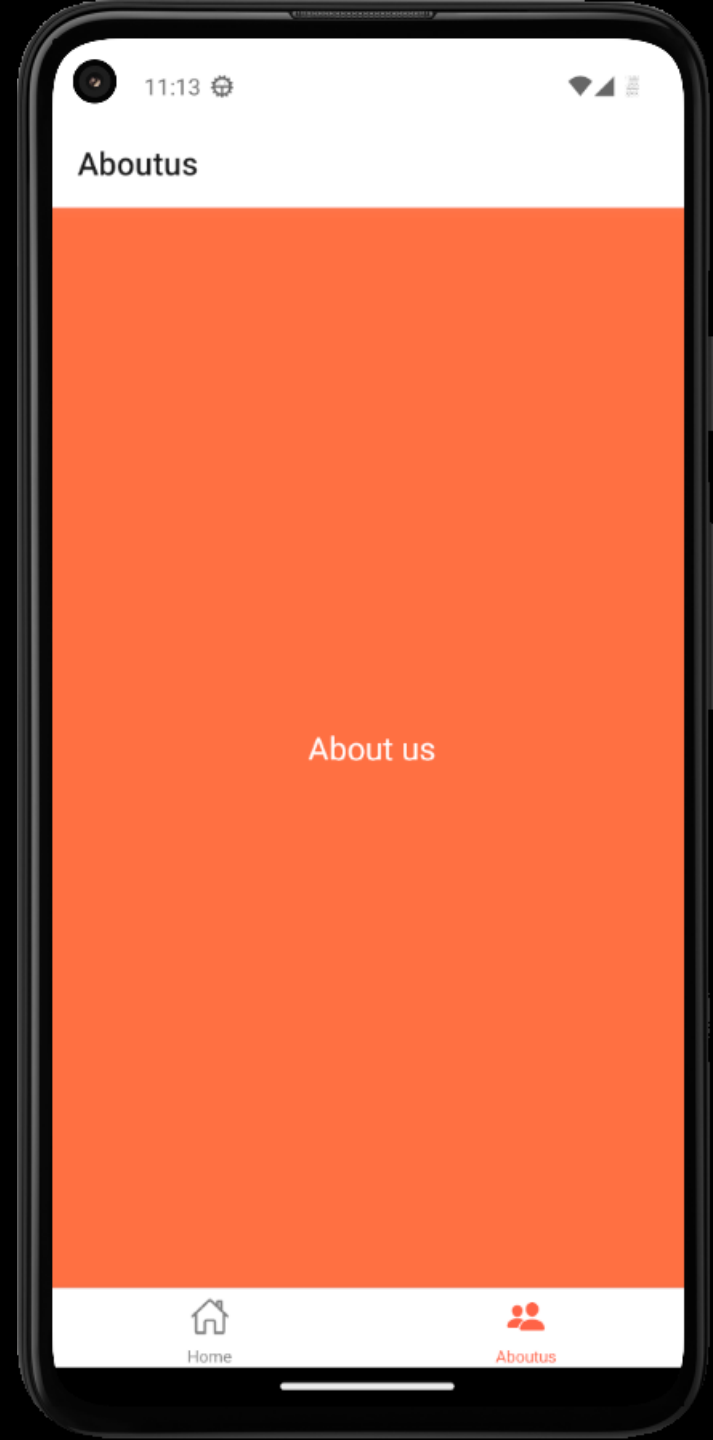
- Create Tab navigator

```
const Tab = createBottomTabNavigator();
```

```
const Screen1 = ({navigation}) => {  
  return (  
    <View style={styles.screen1}>  
      <Image source={require("../assets/img1.jpg")}  
        style={styles.img}/>  
      <Text style={styles.title}>  
        Welcome to my app    </Text>  
    </View>  
  );  
}
```



```
const Screen2 = ({navigation}) =>{  
  return (  
    <View style={styles.screen2}>  
      <Text style={styles.title}> About us</Text>  
  
    </View>  
  );  
}
```



```
const Tab = createBottomTabNavigator();
const TabNav = () =>
{
  return (
    <NavigationContainer>
      <Tab.Navigator screenOptions={({ route }) => ({
        tabBarIcon: ({ focused, color, size }) => {
          let iconName;
          if (route.name === 'Home') {
            iconName = focused ? 'home' : 'home-outline';
          } else if (route.name === 'Aboutus') {
            iconName = focused ? 'people' : 'people-outline';
          }

          return <Icons name={iconName} size={size} color={color} />;
        },
        tabBarActiveTintColor: 'tomato',
        tabBarInactiveTintColor: 'gray',
      })} >
        <Tab.Screen name="Home" component={Screen1} />
        <Tab.Screen name="Aboutus" component={Screen2} />
      </Tab.Navigator>
    </NavigationContainer>
  )
}
```